Incomplete Information in Production Networks

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Yale

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Motivation: The Nature of Transactions in Production Networks

- Large network literature: macro consequences of micro shocks

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Q: How do micro shocks aggregate to economy-wide disturbances when firms transact under incomplete information?

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- Aggregate impact of sectoral shocks systematically changes in recessions

  e.g., effect of *aggregate* productivity shock on GDP 50% lower in Covid than complete info

- *Networks and the Macroeconomy* – Acemoglu et al. (2012), Baqaee and Farhi (2019, 2020), Carvalho and Tahbaz-Salehi (2019, 2020), Liu (2019), many others...
- Input-Output Linkages with Uncertainty – Atolia and Chahrour ('20), Chahrour, Nimark, and Pitschner ('21), Tian ('21), La'O and Tahbaz-Salehi ('22), Pellet and Tahbaz-Salehi ('23), Bui et al. ('24), Fang et al. ('24), Kopytov et al. (2024), Castro-Vincenzi et al. ('26) ...

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- *Expectations and the Business Cycle* – Mankiw and Reis (2002), Lorenzoni (2009), Jurado, Ludvigson, and Ng (2015), Angeletos and Lian (2018)...

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- *Network Games of Incomplete Information* – Marschak (1955), Calvó-Armengol, de Martí, and Prat (2015), de Martí and Zenou (2015), Bergemann, Heumann, and Morris (2017), Lian (2021), Erol and Vohra (2022a,b)...

Model

Theoretical Results

Quantification

Conclusion

1. Model

Modelling Input-Output Linkages

- Time t is discrete and infinite, but firms make static decisions
- Sectors indexed by $n \in \mathcal{N} = \{1, 2, \ldots, N\}$ with competitive firms
- Each firm produces with CRS Cobb-Douglas Technology (N inputs + labor)

$$q_{nt} = c_n \times z_{nt} \times L_{nt}^{\alpha_{nl}} \prod_{n' \in \mathcal{N}} X_{nn't}^{\alpha_{nn'}}, \quad \alpha_{nn'} \in [0, 1), \, \alpha_{nl} > 0$$

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- Define *input-output matrix*: $\mathbf{A} = [\alpha_{nn'}]$, *labor-share matrix*: $\mathbf{A}^l = \mathrm{diag}(\alpha_{nl})$

Representative Household Decision Rules

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- Decision rules can be microfounded w/ Golosov-Lucas ('07) preferences Details

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Key assumption: Firms don't observe $\{z_{nt}\}_{n \in \mathcal{N}}$, M_t *directly*, but can condition input purchases on subset of *market outcomes* ($\mathcal{I}_{nn't} \subseteq \{p_{nt}\}_{n \in \mathcal{N}}$) Contracting

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- Info structure: How flexibly inputs can **respond** to different **economic conditions**
- Labor adjusts to realized demand (*e.g.*, overtime work, capacity utilization...)

Rational Expectations Equilibrium

Definition (REE)

An equilibrium is a *collection* of variables

$$\{\{\{X_{nn't}\}_{n'\in\mathcal{N}}\}, p_{nt}, C_{nt}, L_{nt}, q_{nt}, z_{nt}\}_{n\in\mathcal{N}}, \mathcal{C}_t, \mathcal{P}_t, L_t, w_t, M_t\}_{t\in\mathbb{N}}$$

such that:

1. Households maximize expected utility + supply labor to meet demand
2. Firms optimize given available information:

$$L_{nt} = \arg\max_{L_{nt}} \Pi_{nt} \qquad X_{nn't} = \arg\max_{X_{nn't}} \mathbb{E}_t\left[(\mathcal{P}_t\mathcal{C}_t)^{-1}\Pi_{nt}|\mathcal{I}_{nn't}\right]$$

3. Expectations are consistent with the equilibrium laws of motion and markets clear

2. Theoretical Results

Warm-Up: Frictionless Networks and Domar Weights

- Firms choose:

$$L_{nt} = \alpha_{nl} \frac{R_{nt}}{w_t} \quad \text{and} \quad X_{nn't} = \alpha_{nn'} \frac{R_{nt}}{p_{n't}}$$

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- Define $\hat{x}_t = \log x_t$

Real GDP in the Frictionless Benchmark

$$\hat{C}_t = cons + \lambda' \hat{z}_t + \chi \hat{M}_t$$

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- Look at sector's Domar weight ("size") $\rightarrow$ impact of shock

Introducing Uncertainty: Firms Respond to Prices Differently

Demand Functions under Incomplete Information

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- **Point**: optimal choice of inputs depends on how well firms can *forecast* revenues
- Dual role of prices as in Lucas (1972) or Grossman and Stiglitz (1980)

Illustration: A Simple Vertical Supply Chain with Incomplete Info

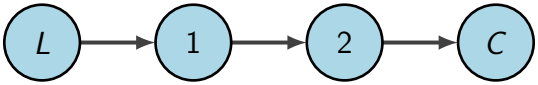
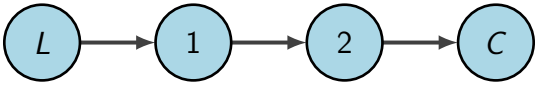
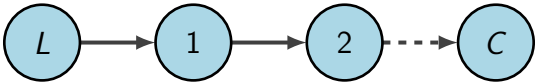


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- Firm 2's revenues must be equal to total expenditures: $\hat{R}_{2t} = \hat{M}_t$

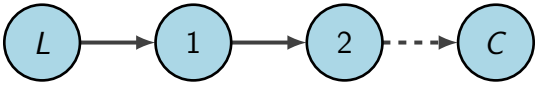
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- Firm 1's price is equal to marginal cost:

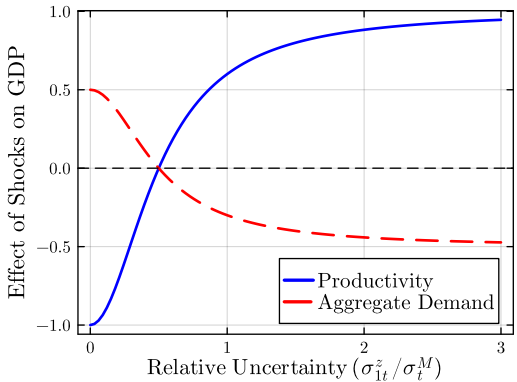
$$\hat{p}_{1t} = cons - \overbrace{\hat{z}_{1t}}^{\text{productivity}} + \overbrace{(1-\chi)\hat{M}_t}^{\text{wage}}$$

Real GDP in the Simple Vertical Supply Chain

$$\hat{\mathcal{C}}_t = cons + \left(-\frac{\mathrm{d}\log X_{21,t}}{\mathrm{d}\log p_{1t}}\right) \underbrace{\left[\hat{z}_{1t} - (1-\chi)\hat{M}_t\right]}_{-\log p_{1t}}$$

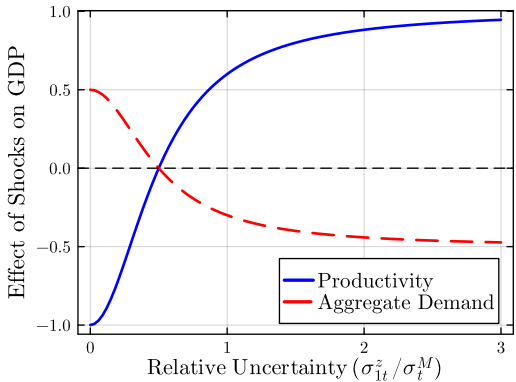
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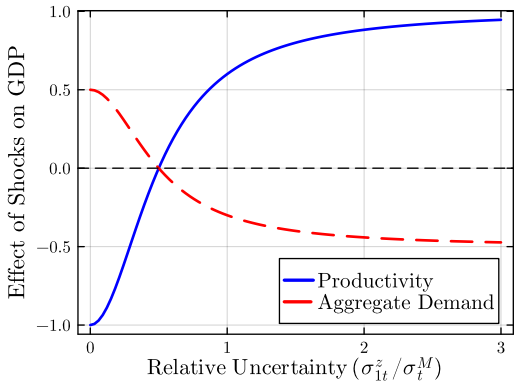
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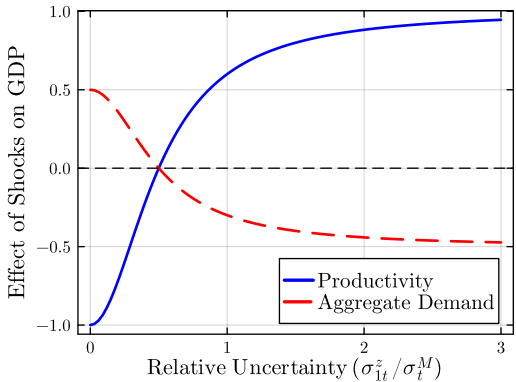
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Equilibrium Characterization for General Network Structures

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$$R_{nt} = z_{nt} p_{nt} \left(\frac{R_{nt}}{M_t^{1-\chi}}\right)^{\alpha_{nl}} \prod_{n' \in \mathcal{N}} \left(\frac{\mathbb{E}_t\left[M_t^{-1} R_{nt} \middle| \mathcal{I}_{nn't}\right]}{\mathbb{E}_t\left[M_t^{-1} p_{n't} \middle| \mathcal{I}_{nn't}\right]}\right)^{\alpha_{nn'}}$$

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- **Key challenge**: choices depend on stochastic properties of *endogenous* objects

Responsiveness and Outcomes: A Two-Step Procedure

- Use **two-step procedure**:
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$$\mathbf{\Omega}_t = (\omega_{nkt}^{n'})_{n,k,n'=1}^{N} = \left(-\frac{\mathrm{d} \log X_{nkt}}{\mathrm{d} \log p_{n't}} \right)_{n,k,n'=1}^{N} \qquad \text{(elasticity of } n \text{ to } k \text{ due to } n')$$

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- Also define revenue shares $[s_{nn'}]$ (sales share of n to n')

A Representation of Supply and Demand under Incomplete Info

Proposition (Network Demands and Supplies)

$$\log q_t^d = cons - \mathcal{E}_t^D \log p_t + \mathrm{diag}(s_{nc}) \log M_t$$
$$(\mathbf{I} - \mathbf{A}^I) \log q_t^s = cons + \mathcal{E}_t^S \log p_t + [\log z_t - (1-\chi)\mathbf{A}^I \log M_t]$$

where $\mathcal{E}_t^D$ *and* $\mathcal{E}_t^S$ *are given by*

$$\left[\mathcal{E}_t^D\right]_{nn'} = s_{nc}\, 1\{n = n'\} + \sum_{k \in \mathcal{N}} s_{nk}\, \omega_{kn,t}^{n'}, \quad \left[\mathcal{E}_t^S\right]_{nn'} = \alpha_{nl}\, 1\{n = n'\} - \sum_{k \in \mathcal{N}} \alpha_{nk}\, \omega_{nk,t}^{n'}$$

- No equivalence under complete info w/ CRS (no substitution theorem)

The Aggregate Impact of Shocks Depends on Responsiveness and Technology

Theorem (The Economy's Adjusted-by-Uncertainty Domar Index (AUDI))

$$\frac{\mathrm{d}\log \mathcal{C}_t}{\mathrm{d}\log z_t} \equiv \epsilon(\mathbf{\Omega}_t)' = \gamma'\left(\mathcal{E}_t^S + (\mathbf{I} - \mathbf{A}^l)\mathcal{E}_t^D\right)^{-1}$$

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- Tâtonnement Process:

$$\mathbf{I} + (\mathbf{I} - \mathcal{H}_t) + (\mathbf{I} - \mathcal{H}_t)^2 + \dots$$

  where $\mathcal{H}_t \equiv \left(\mathcal{E}_t^S + (\mathbf{I} - \mathbf{A}^I)\mathcal{E}_t^D\right)$

- Each step is an iteration of excess demand: $(\mathbf{I} - \mathbf{A}^I)(\hat{q}_t^d - \hat{q}_t^s) = -\hat{z}_t - \mathcal{H}_t\hat{p}_t$

How Does this Differ From the Complete Info Benchmark?

- If $\omega_{nk,t}^{n'} = 1\{k = n'\} \implies \mathcal{H}_t = \mathbf{I} - \mathbf{A}$
- Own-price elasticities of one (and cross-elasticities of zero) recover Leontief inverse

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- Consider cost-contingent info structure with $\omega^{n'}_{nk,t} = \omega^*_t \times 1\{k = n'\}$:

$$\epsilon(\mathbf{\Omega}_t)' = \gamma' \left[\sum_{j=0}^{\infty} \left(\omega^*_t \mathbf{D}_t \mathbf{A} \right)^j \right] \mathbf{D}_t \quad \text{with} \quad \mathbf{D}_t \equiv \text{diag}\left(\frac{1}{1 - (1 - \alpha_{nl})(1 - s_{nc})(1 - \omega^*_t)} \right)$$

- Higher-order linkages attenuated by responsiveness (alpha centrality)

How Does Responsiveness Affect Effective Network Structure?

- Consider perturbation around frictionless benchmark: $\epsilon(\mathbf{\Omega}_t^*)' = \gamma'(\mathbf{I} - \mathbf{A})^{-1}$

Proposition (Responsiveness and AUDI Weights)

The AUDI weights of all sectors are increasing in any $\omega_{nkt}^{n'}$ around $\mathbf{\Omega}_t^*$. Moreover:

$$\frac{\mathrm{d}\epsilon(\mathbf{\Omega}_t)'}{\mathrm{d}\omega_{nkt}^{n'}}\Big|_{\mathbf{\Omega_t}=\mathbf{\Omega_t^*}} = \alpha_{nk}\alpha_{kl}\lambda_n \left[e'_{n'}(\mathbf{I} - \mathbf{A})^{-1}\right]$$

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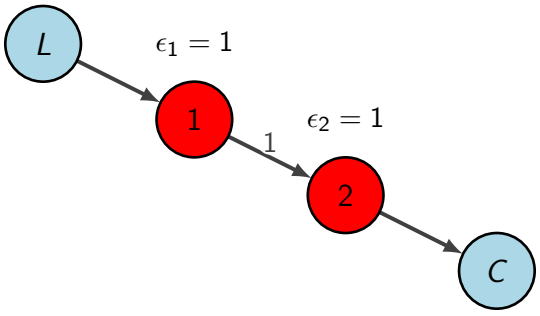
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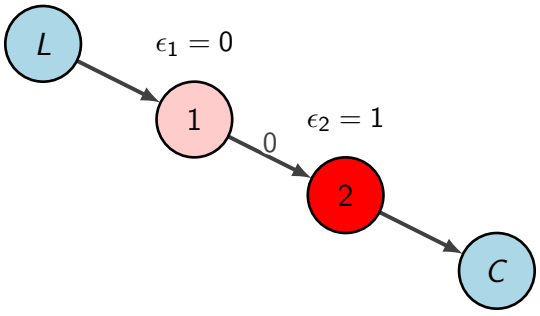
- $\left[e_{n'}'(\mathbf{I} - \mathbf{A})^{-1}\right]_n$ is **upstreamness** of n wrt to n': $\sum_{j=0}^{\infty}(\mathbf{A}^j)_{n'n}$
- Responsiveness primarily matters for *upstream* sectors!

Illustration of AUDI: Back to the Vertical Supply Chain



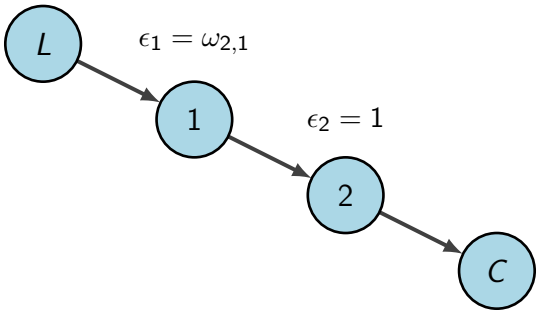
The centrality of the **upstream** sector is **high** when customer's responsiveness is high

Illustration of AUDI: Back to the Vertical Supply Chain



The centrality of the **upstream** sector is **low** when customer's responsiveness is low

Illustration of AUDI: Back to the Vertical Supply Chain

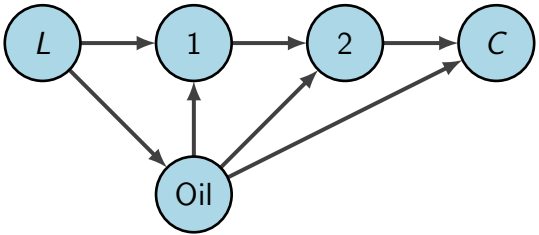


Simple Chain: Centrality = Responsiveness of Customer

An "Oil" Network: Illustrating the Role of Higher-Order Linkages

- An "Oil" network with equal input shares: $\alpha_{nn'} = \gamma_{n'} = \frac{1}{2}$ for all n, n'

$$\epsilon_1 = \frac{1}{2}\frac{\omega_{2,1}}{1+\omega_{2,1}} \qquad \epsilon_2 = \frac{1}{2}$$

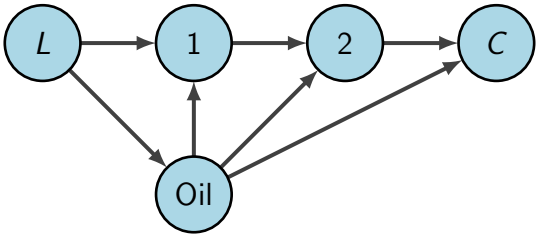


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"Importance" of oil matters through higher-order linkages

(responsiveness of customers' customer)

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- *e.g.*, under *cost-contingent info* structure ($\mathcal{I}_{nn't} = \{p_{n't}\}$):

$$[\omega^{n'}_{nn't}] = \left[1 - \frac{\mathrm{Cov}(\log R_{nt}, \log p_{n't})}{\mathrm{Var}(\log p_{n't})}\right]$$

Input Responsiveness under Dominant Productivity Uncertainty

- Define *uncertainty ratio* of productivity to demand:

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Theorem (High Productivity Uncertainty)

Suppose $P_{kt} \in \mathcal{I}_{nk,t}$ *for all* $n, k \in \mathcal{N}$. *Let* $u_{nt} \to \infty$ *for all* $n \in \mathcal{N}$.

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- **Point**: *High* productivity uncertainty $\to$ *low* impact of info frictions...
- Theorem can be relaxed under suitable rank condition

Details

Input Responsiveness under Dominant Demand Uncertainty

Theorem (High Demand Uncertainty)

Suppose $\mathcal{I}_{nk} \neq \emptyset$ *for all* $n, k \in \mathcal{N}$ *with* $\alpha_{nk} > 0$. *If* $u_{nt} \to 0$ *for all* $n \in \mathcal{N}$, *then*

$$\sum_{n' \in \mathcal{I}_{nk}} \omega_{nk,t}^{n'} = -\frac{\chi}{1-\chi}, \quad \frac{\mathrm{d} \log C_t}{\mathrm{d} \log M_t} = \chi$$

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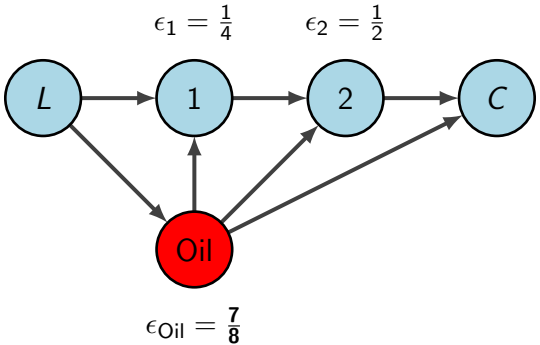
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- *e.g.*, under cost-contingent info, *all* input responsiveness is zero with $\chi = 0$
- **Point**: *High* demand uncertainty $\to$ *attenuation* of input-output linkages

Back to the Oil Network: The Role of Relative Uncertainty

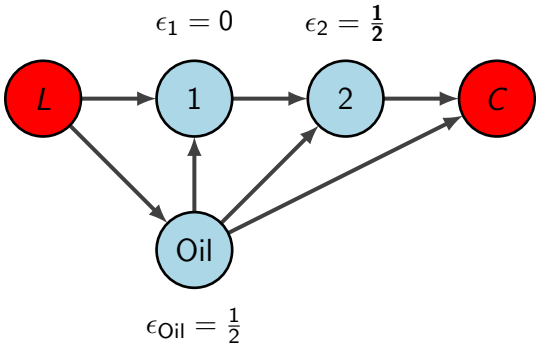
Lots of Supply Shocks in Oil Sector: $\sigma^z_{Oil,t} \gg \sigma^M_t$



- AUDI = Domar for *all* sectors

Back to the Oil Network: The Role of Relative Uncertainty

Lots of Aggregate Demand Uncertainty: $\sigma_t^M \gg \sigma_{\text{Oil},t}^z,\ \sigma_{1,t}^z$ (and $\chi = 0$)



- Only *direct* linkages to households matter $\rightarrow$ downstream sectors more important!

Networks + Uncertainty: Key Theoretical Takeaways

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2. **Uncertainty**: Info frictions change responsiveness (relative to complete info)
   Depends on relative uncertainty about productivity *vs.* aggregate demand

3. **Aggregation**: Domar weights poor approx. (esp. when high demand uncertainty)
   Inco-Info qualitatively changes pass-through of shocks (impact can even flip sign)

3. Quantification

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- **Need**: (i) Input-output structure; (ii) proxies for uncertainty
- **Data**:
  1. BEA input-output tables (2022) $\rightarrow$ network structure (66 sectors)
  2. ILPA sectoral productivity data (1987 - 2023) $\rightarrow z_{nt}$
  3. Nominal GDP growth $\rightarrow M_t$

Estimating Time-Varying Uncertainty: GARCH Model

- Estimate $Z_t = \{\sigma^z_{nt}\epsilon^z_{nt}\}_{n\in\mathcal{N}}$ and $Z_t = \sigma^M_t\epsilon^M_t$ using CCC GARCH(1,1)

$$Z_t \sim N(0, \Sigma_t), \qquad \Sigma_t = D_t^{\frac{1}{2}} C D_t^{\frac{1}{2}}$$

- D_t diagonal matrix with entries:

$$\sigma^2_{i,t} = s_i + \alpha_i \varepsilon^2_{i,t-1} + \beta_i \sigma^2_{i,t-1}$$

Estimating Time-Varying Uncertainty: GARCH Model

- Estimate $Z_t = \{\sigma^z_{nt}\epsilon^z_{nt}\}_{n\in\mathcal{N}}$ and $Z_t = \sigma^M_t\epsilon^M_t$ using CCC GARCH(1,1)

$$Z_t \sim N(0, \Sigma_t), \qquad \Sigma_t = D_t^{\frac{1}{2}} C D_t^{\frac{1}{2}}$$

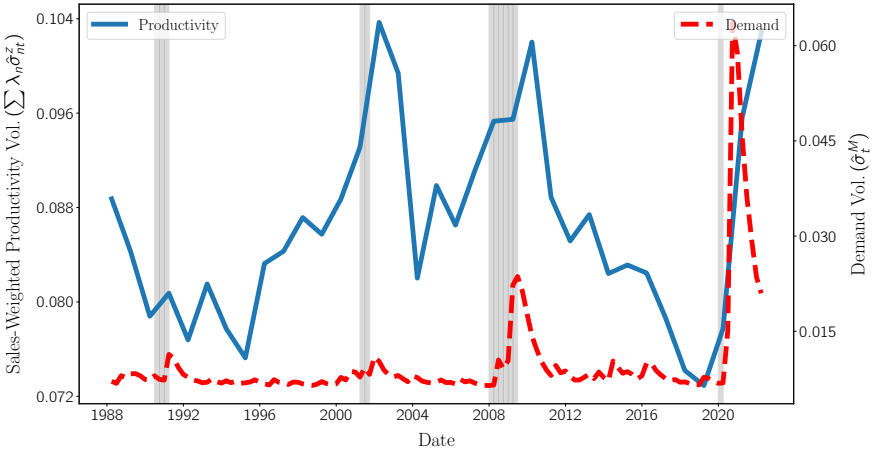
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- Obtain time-varying estimates: $\hat{\sigma}^z_{nt}$, $\hat{\sigma}^M_t$

- Set $\chi = 0.91$ to match covariance of nominal wages with nominal GDP

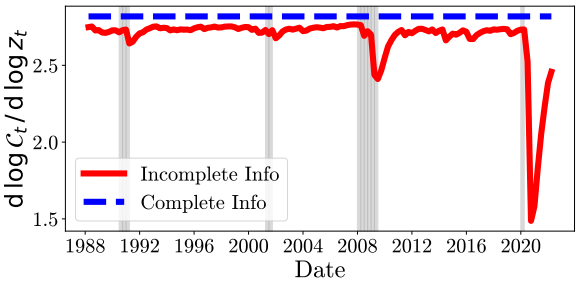
Both Productivity and Demand Volatility Spike in Recessions



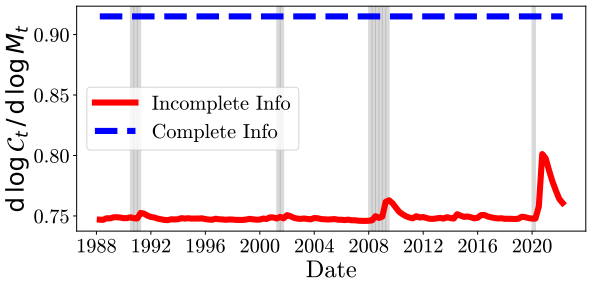
Focus has been on **absolute** vols (*e.g.*, Bloom et al., 2018), but **relative** vols matter

Uncertainty + Network → State-Dependent Impact of Aggregate Shocks

(a) Aggregate Productivity Shock

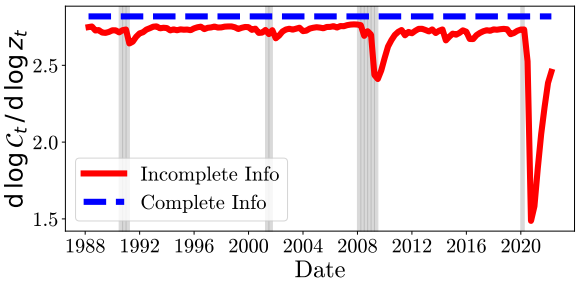


(b) Aggregate Demand Shock

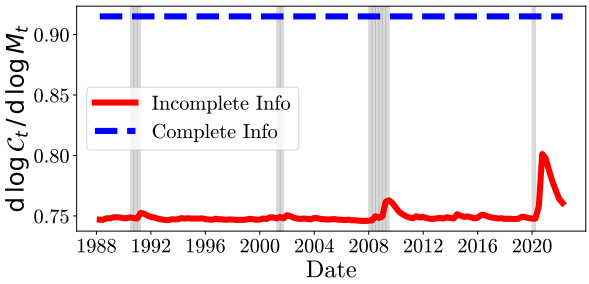


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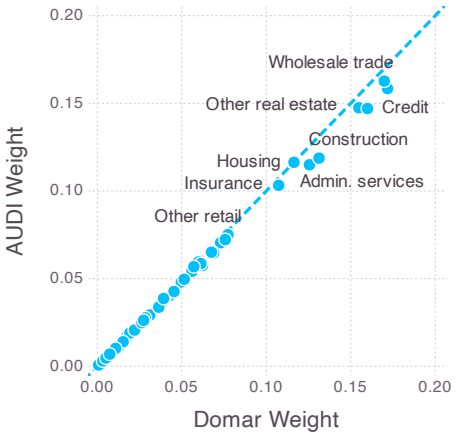


Variance of Output Growth (1987-2023) Relative to Data

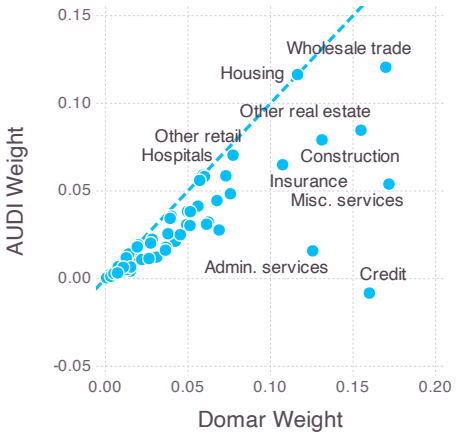
Model	*Complete Info*	*Fixed Responsiveness*
92%	155%	57%

Sectoral Importance in Times of Low and High Uncertainty

(a) Normal Times



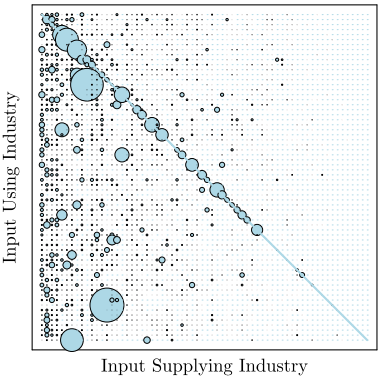
(b) Covid



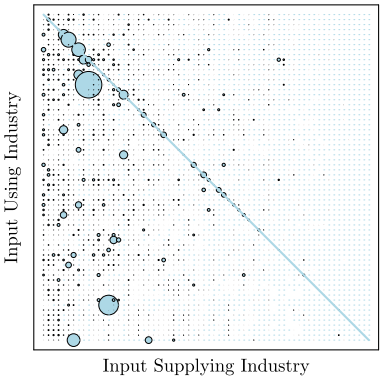
Changes in Responsiveness Drives Changes in Sectoral Importance

The Economy's Input Demand Matrices: $\mathbf{A} \odot \mathbf{\Omega}_t$

(a) Normal Times



(b) Covid



Upstream Sectors Matter More when Productivity Uncertainty is High

Sector	% Change in AUDI: $\epsilon(\hat{\mathbf{\Omega}}_t)/\epsilon(\hat{\mathbf{\Omega}}_{Normal}) - 1$			Upstreamness
	Dot Com	*Financial Crisis*	*Covid*	
Miscellaneous services	0.01	-0.14	-0.66	8.67
Wholesale trade	0.00	-0.10	-0.26	9.63
Credit intermediation	0.03	-0.08	-1.06	6.60
Other real estate	0.00	-0.11	-0.43	9.36
Construction	0.00	-0.13	-0.33	2.71
Admin. & support services	0.01	-0.23	-0.86	9.16
Housing	0.00	0.00	0.00	1.00
Insurance carriers	0.01	-0.10	-0.37	7.03
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4. Conclusion

Informational Frictions Matter for the "Chain of Transactions"

Canonical assumption: trading under complete and contingent contracts

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2. State-dependence: changes in uncertainty $\Longrightarrow$ changes in network structure

Informational Frictions Matter for the "Chain of Transactions"

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This paper: frictional transactions due to incomplete information

1. Effective network structure mediated by incomplete information
2. State-dependence: changes in uncertainty $\Longrightarrow$ changes in network structure
3. Who matters?
   - "Supply" uncertainty: upstream sectors
   - "Demand" uncertainty: downstream sectors

Methodologically: *network interactions* + learning from *endogenous* objects

Thank You!

georgios.nikolakoudis@yale.edu

A Contracting Interpretation to the Informational Structure

- Equivalence between "incomplete contracting" and "incomplete information"
- Supplier and producer bargain over transfer $\tau(Z)$ and input delivery $X(Z)$
  - Z: variables contract is contingent on
- Common contract: $Z =$ supplier's cost of making input ($\mathcal{M}$)
  - "Cost-plus contracts": Fisher Body and General Motors, construction (Bajari and Tadelis, 2001), software (Banerjee and Duflo 2000), defense (Hiller and Tollison, 1973), transportation (Crocker and Reynolds, 1993) ...

A Contracting Interpretation to the Informational Structure

- Nash bargaining over contract

$$\max_{\tau(\mathcal{M}),x(\mathcal{M})} \left(\mathbb{E}_t\left[R(x(\mathcal{M}))\right] - \mathbb{E}_t\left[\tau(\mathcal{M})\right]\right)^{\beta} \left(\mathbb{E}_t\left[\tau(\mathcal{M})\right] - \mathbb{E}_t[\mathcal{M}]\right)^{1-\beta}$$

  where $R(\mathcal{M})$: *producer*'s revenues

- **In paper**:
  1. Solve for deliveries in optimal contract $x(\mathcal{M})$ using variational techniques
  2. Show that it implements same demand function for inputs as model (with $\mathcal{M} = P_{n't}$ for relevant sector)

- **Intuition**: "Cost plus" contracts: same measurability constraint on input choices

Back

Representative Household: Assets and Payoffs

- Traded assets: *money* M_t and short-term risk-free *bonds* B_t

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$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t\left(\log \mathcal{C}_t + \log\frac{M_t}{\mathcal{P}_t} - L_t\right)\right], \quad \log \mathcal{C}_t = \sum_{n\in\mathcal{N}}\gamma_n \log C_{nt}$$

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- Wages: $w_t = (w_{t-1})^{\chi}(w_t^*)^{1-\chi}$, where $\chi \in [0,1)$ back

AUDI: The Macro Impact of Micro Shocks under Uncertainty

Back

Theorem (The Economy's Adjusted-by-Uncertainty Domar Index)

Given $\mathbf{\Omega}_t$, the first-order impact of shocks on real GDP satisfies:

$$\hat{\mathcal{C}}_t = \epsilon(\mathbf{\Omega}_t)'\hat{z}_t + \left[1 - \epsilon(\mathbf{\Omega}_t)' \times \operatorname{diag}\left((1-\chi)\alpha_{nl} + (1-\alpha_{nl})s_{nc}\right)\right] \hat{M}_t$$

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- Suppose $\omega_{nn't} = \omega_t^*$. Then,

$$\epsilon(\omega_t^*)' = \gamma' \left[\sum_{k=0}^{\infty} \Big(\underbrace{\omega_t^* \left(\mathbf{I} - \operatorname{diag}(b_{nt})\right)^{-1}}_{\text{input-output weights}} \Big)^k \mathbf{A}^k \right] \left(\mathbf{I} - \operatorname{diag}(b_{nt})\right)^{-1}$$

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- Higher-order linkages are weighted by ω_t^* — **alpha centrality** (Katz, 1953)

Incomplete Information Changes the Network's Alpha Centrality

Back

- Sector's "alpha centrality" defined as

$$AC = \boldsymbol{\gamma}'(\mathbf{I} - \alpha\mathbf{A})^{-1} = \boldsymbol{\gamma}' \sum_{k=0}^{\infty} \alpha^k \mathbf{A}^k$$

connections to distant sectors "penalized" by factor $\alpha \in (0, 1)$

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- Alpha centrality is endogenous and shaped by responsiveness

Input Responsiveness: Firms Use Prices to Predict their Revenues

Proposition (Optimal Responsiveness)

In all log-linear equilibria, input responsiveness is given by:

$$\mathbf{\Omega}_t = [\omega_{nn't}] = \left[1 - \frac{Cov(\hat{R}_{nt}, \hat{p}_{n't})}{Var(\hat{p}_{n't})}\right]$$

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2. *High* cost variation $\rightarrow$ *high* responsiveness

Back

Conditioning on Further Input Prices: Equilibrium Characterization

- When purchasing input n: firms observe other input prices *and* price of input n
- $\mathcal{I}_{nt}$: other prices observable to firms in sector n (does not include wage)

$$R_{nt} = \tilde{c}_{nt} z_{nt} P_{nt} \left(\frac{R_{nt}}{M_t^{1-\chi}} \right)^{\alpha_{nl}} \prod_{n' \in \mathcal{N}} \left(\frac{\mathbb{E}_t \left[M_t^{-1} R_{nt} \middle| P_{n't}, \mathcal{I}_{nt} \right]}{\mathbb{E}_t \left[M_t^{-1} P_{n't} \middle| P_{n't}, \mathcal{I}_{nt} \right]} \right)^{\alpha_{nn'}}$$

$$R_{nt} = \gamma_n \iota_t M_t + \sum_{n' \in \mathcal{N}} \alpha_{n'n} P_{nt} \frac{\mathbb{E}_t \left[M_t^{-1} R_{n't} \middle| P_{n't}, \mathcal{I}_{n't} \right]}{\mathbb{E}_t \left[M_t^{-1} P_{nt} \middle| P_{n't}, \mathcal{I}_{n't} \right]}$$

where $\tilde{c}_t = (\iota_t)^{\alpha_{nl}(1-\chi)} (w_{t-1})^{\alpha_{nl}\chi}$

Equilibrium with Exogenous Public Signals

- Exogenous public signal given by:

$$\log s_t^p = \log M_t + \varepsilon_t^p$$

- Signal-to-noise ratio

$$\kappa_t = \frac{(\sigma_t^M)^2}{(\sigma_t^M)^2 + (\sigma_t^p)^2}$$

Proposition

Real GDP under fixed responsiveness is given by:

$$\log C_t = cons + \epsilon(\mathbf{\Omega}_t)' \log z_t + [1 - \gamma' \mathrm{diag}(a_n)] \log M_t - \gamma' \mathrm{diag}(b_n) \log s_t^p$$

- AUDI: same as before (but coefficients on other shocks different)

Responsiveness with an Exogenous Public Signal

Lemma

There exists an equilibrium in which input responsiveness is given by:

$$\mathbf{\Omega}_t = [\omega_{nn't}] = \left[1 - \frac{Cov(\log R_{nt}, \log P_{n't} | s_t^p)}{Var(\log P_{n't} | s_t^p)}\right] \tag{1}$$

$$= \left[1 - \frac{Cov(\log R_{nt}, \log P_{n't}) - \frac{Cov(\log R_{nt}, \log s_t^p) Cov(\log P_{n't}, \log s_t^p)}{Var(\log s_t^p)}}{Var(P_{n't}) - \frac{Cov^2(\log P_{n't}, \log s_t^p)}{Var(\log s_t^p)}}\right] \tag{2}$$

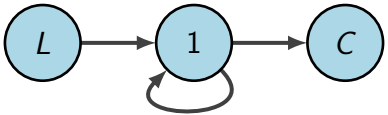
- Optimal responsiveness:

$$\log X^*_{nn't} = \beta_0 + \beta_1 \log P_{n't} + \beta_2 \sigma_t^p + \text{error}$$

Back

Roundabout Economy

Back

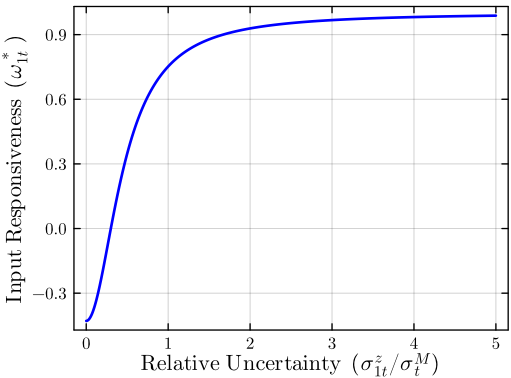


Proposition

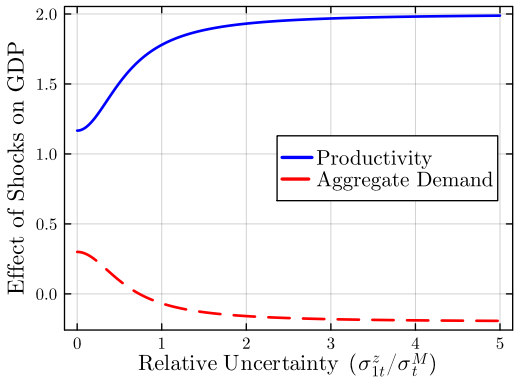
Optimal responsiveness ω_t^* in the roundabout economy is given by:

$$\omega_t^* = 1 - \frac{1}{1 + \frac{(1-\alpha_1)^{-2}}{1+\alpha_1-\chi} u_{1t}^2 - \chi} \tag{3}$$

(a) Input Responsiveness

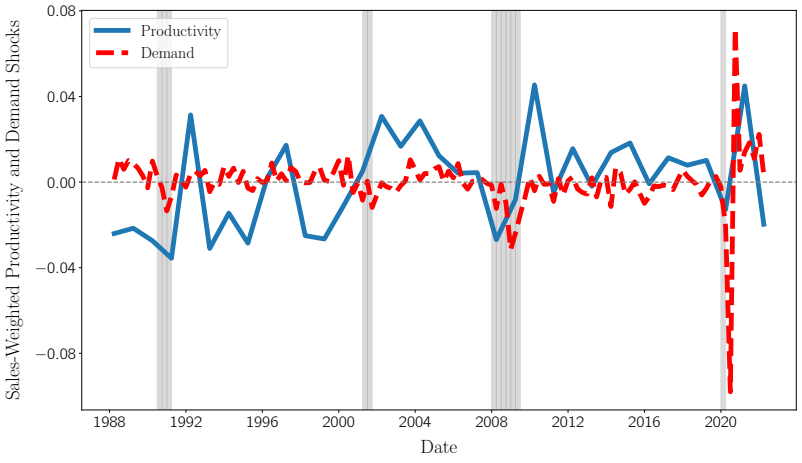


(b) Effect of Shocks on GDP



Estimated Productivity and Demand Shocks

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Theorem (Dominant Uncertainty Limits)

Suppose $P_{kt} \in \mathcal{I}_{nk}$ *for all* $n, k \in \mathcal{U} \subseteq \mathcal{N}$. *Let* $u_{nt} \to \infty$ *for all* $n \in \mathcal{U} \subseteq \mathcal{N}$. *If,* $\mathbf{1} \in \operatorname{col}([(\mathbf{I} - \mathbf{A})^{-1}]_{\mathcal{I}_{nk}, \mathcal{U}})$ *for all* $n, k \in \mathcal{N}$,

$$\omega_{nk,t}^{n'} = 1\{n' = k\}, \qquad \frac{\mathrm{d} \log \mathcal{C}_t}{\mathrm{d} \log z_t} = \gamma'(\mathbf{I} - \mathbf{A})^{-1}$$

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Why Prices? Contracts Respond Only to What They Are Written On

- Bargain over $x(Z), \tau(Z)$ contingent on *verifiable* Z (model with $\mathcal{I}_{nkt} = Z$)
- Info structure = contract's **form** (what it conditions on)

Info set $\mathcal{I}_{nkt}$		**Contract form**	**Where we see it**
commitment	$\emptyset$	fixed quantity, take-or-pay	natural gas, coal (Masten–Crocker '85; Joskow '85)
cost-contingent	$\{P_{kt}\}$	cost-plus, cost escalation	coal, construction, defense, software (Joskow '88; Bajari–Tadelis '01)
key-input contingent	$\{P_{jt}\}_{j \in \mathcal{K}}$	surcharges, index escalation	fuel surcharges (trucking, airlines), steel surcharges (autos), PPI clauses (BLS '21)
demand-contingent	$\{P_{nt}\}$	netback, revenue sharing	gas netback pricing, retail (Cachon–Lariviere '05)
complete info	full state	Arrow–Debreu contingent	not observed: costly to write (Battigalli–Maggi '02)

Bargaining Back